In-Person Worksheet: Unit 2

Name: _____ **Date:** _____

Easy problem first. If you get it, try the hard one.

2.1–2.2 Sequences and change

1E. The sequence $7, 11, 15, 19, ...$ is arithmetic or geometric? Write a rule for the nth term and find the 10th term.

1H. A geometric sequence has $a_2 = 12$ and $a_5 = 324$.

- (a) Find the common ratio and a_1.
- (b) Write a_n as an exponential function of n.
- (c) Find the average rate of change on $[1, 2]$ and on $[2, 3]$. Why isn't it constant, even though the ratio is?

2.3 Exponential functions

2E. Find the exponential function through $(0, 6)$ and $(2, 54)$. Growth or decay, and by what percent per step?

2H. $g(x) = -2 \cdot 3^{x-1} + 4$

- (a) Horizontal asymptote? (b) End behavior as $x \to \infty$ and $x \to -\infty$.
- (c) y-intercept? (d) Range? Is it concave up or down?

2.4 Exponential models

3E. \$1200 is invested at 5% interest compounded annually. Find the value after 8 years. *(Calculator OK)*

3H. A car worth \$24,000 loses 15% of its value each year.

- (a) Write a model. (b) When is it worth less than \$10,000? *(Calculator OK)*
- (c) Compare to a car that loses a **constant** \$3,000 per year. Who is worth more after 5 years?

2.6 Composition

4E. $f(x) = x + 3$ and $g(x) = 2x^2$. Find $f(g(1))$ and $g(f(1))$.

4H. $f(x) = 2^x$ and $g(x) = x^2 - 1$.

- (a) Find $f(g(x))$ and $g(f(x))$. (b) Evaluate both at $x = 3$.
- (c) Which one is an exponential function? Explain.

2.7 Inverse functions

5E. Find $f^{-1}(x)$ for $f(x) = 5x - 10$, and find $f^{-1}(15)$.

5H. $f(x) = \dfrac{3x}{x-2}$

- (a) Find $f^{-1}(x)$. (b) Domain of f and of f^{-1}.
- (c) $f(4) = 6$. Use your inverse to confirm $f^{-1}(6) = 4$.

2.8–2.9 Logarithms and inverses of exponentials

6E. Evaluate without a calculator: (a) $\log_3 81$ (b) $\log 0.01$ (c) $\ln 1$ (d) $\log_4 2$

6H. $f(x) = 2^x + 1$

- (a) Find $f^{-1}(x)$. (b) Domain and range of f and of f^{-1}.
- (c) Find $f^{-1}(9)$. (d) How are the graphs of f and f^{-1} related?

2.10 Logarithmic functions

7E. $g(x) = \log_2 x$. State the domain, the vertical asymptote, the x-intercept, and $g(8)$.

7H. $h(x) = -\log_3(x+1) + 2$

- (a) Describe the transformations from $\log_3 x$. (b) Domain and vertical asymptote.
- (c) Find $h(2)$ and the zero of h. (d) End behavior at both ends of the domain.

2.11 Properties of logarithms

8E. Write as a single logarithm: $\ln 6 + \ln 4 - \ln 8$.

8H. Suppose $\log_b 2 = 0.35$ and $\log_b 3 = 0.55$.

- (a) Find $\log_b 18$. (b) Find $\log_b \frac{4}{27}$. (c) Find $\log_b \sqrt{6}$.
- (d) A student wrote $\log(x+3) = \log x + \log 3$. Find a value of x that shows it's wrong.

2.12 Exponential and logarithmic equations

9E. Solve $2^{x+1} = 32$.

9H.

- (a) $5 \cdot 3^x = 200$ (exact and decimal)

- (b) $\ln x + \ln(x-4) = \ln 5$
- (c) *Stretch:* $2^{2x} - 5 \cdot 2^x + 4 = 0$ (hint: let $u = 2^x$)

2.13–2.14 Log modeling and semi-log plots

10E. An earthquake's magnitude is $M = \log I$, where I is intensity. If one quake has 1000 times the intensity of another, how much larger is its magnitude?

10H. A table of (x, y): $(0, 5), (1, 15), (2, 45), (3, 135)$.

- (a) Find $\log y$ for each point (calculator OK). (b) Show that $\log y$ versus x is a line, and find its slope.
- (c) Write the line, then convert back to $y = ab^x$.

Answer Key

2.1–2.2 Sequences and change

1E. Constant difference 4, so **arithmetic**: $a_n = 7 + 4(n-1) = 4n + 3$. $a_{10} = 43$.

1H. (a) $a_5/a_2 = r^3 = 27$, so $r = 3$ and $a_1 = 12/3 = 4$. (b) $a_n = 4 \cdot 3^{n-1}$. (c) $a_1, a_2, a_3 = 4, 12, 36$: rates are 8 and 24. The rate triples each time, because an exponential function's change is proportional to its current value. The *ratio* of outputs is constant; the *differences* are not.

2.3 Exponential functions

2E. $a = 6$; $6b^2 = 54$, so $b^2 = 9$ and $b = 3$ (it must be positive): $y = 6 \cdot 3^x$. **Growth**, $3 - 1 = 2$, so **200%** per step.

2H. (a) $y = 4$. (b) As $x \to \infty$, $-2 \cdot 3^{x-1} \to -\infty$, so $g \to -\infty$. As $x \to -\infty$, $3^{x-1} \to 0$, so $g \to 4$. (c) $g(0) = -2 \cdot \frac{1}{3} + 4 = \frac{10}{3}$. (d) Range $y < 4$. The negative stretch flips it: **concave down**.

2.4 Exponential models

3E. $1200(1.05)^8 \approx \$1772.95$.

3H. (a) $V(t) = 24000(0.85)^t$. (b) $0.85^t = \frac{10}{24}$, so $t = \frac{\ln(10/24)}{\ln 0.85} \approx 5.39$ years. (c) Exponential: $24000(0.85)^5 \approx \$10{,}649$. Linear: $24000 - 15000 = \$9000$. The exponential car is worth more, because its loss shrinks as its value shrinks.

2.6 Composition

4E. $g(1) = 2$, so $f(2) = 5$. $f(1) = 4$, so $g(4) = 32$.

4H. (a) $f(g(x)) = 2^{x^2-1}$. $g(f(x)) = (2^x)^2 - 1 = 4^x - 1$. (b) $f(g(3)) = 2^8 = 256$. $g(f(3)) = 64 - 1 = 63$. (c) $g(f(x)) = 4^x - 1$ is a standard exponential (base 4, shifted down). 2^{x^2-1} has x^2 in the exponent, so it is not.

2.7 Inverse functions

5E. $x = 5y - 10$, so $f^{-1}(x) = \dfrac{x + 10}{5}$. $f^{-1}(15) = 5$.

5H. (a) $x = \dfrac{3y}{y-2}$ gives $xy - 2x = 3y$, so $y(x-3) = 2x$ and $f^{-1}(x) = \dfrac{2x}{x-3}$. (b) f: $x \neq 2$. f^{-1}: $x \neq 3$. (c) $f^{-1}(6) = \dfrac{12}{3} = 4$ ✓.

2.8–2.9 Logarithms and inverses of exponentials

6E. (a) 4. (b) -2. (c) 0. (d) $4^{1/2} = 2$, so $\frac{1}{2}$.

6H. (a) $x = 2^y + 1$, so $2^y = x - 1$ and $f^{-1}(x) = \log_2(x-1)$. (b) f: domain all reals, range $y > 1$. f^{-1}: domain $x > 1$, range all reals. (c) $\log_2 8 = 3$. Check: $f(3) = 9$ ✓. (d) Reflections over the line $y = x$; the horizontal asymptote $y = 1$ of f becomes the vertical asymptote $x = 1$ of f^{-1}.

2.10 Logarithmic functions

7E. Domain $x > 0$; VA $x = 0$; x-intercept 1; $g(8) = 3$.

7H. (a) Shift left 1, reflect over the x-axis, shift up 2. (b) $x > -1$; VA $x = -1$. (c) $h(2) = -\log_3 3 + 2 = 1$. Zero: $\log_3(x+1) = 2$, so $x = 8$. (d) As $x \to \infty$, $h \to -\infty$. As $x \to -1^+$, $h \to \infty$.

2.11 Properties of logarithms

8E. $\ln \dfrac{6 \cdot 4}{8} = \ln 3$.

8H. (a) $\log_b(2 \cdot 3^2) = 0.35 + 2(0.55) = 1.45$. (b) $2(0.35) - 3(0.55) = -0.95$. (c) $\frac{1}{2}(0.35 + 0.55) = 0.45$.

(d) $x = 1$: $\log 4 \approx 0.60$ but $\log 1 + \log 3 \approx 0.48$. The rule is for products, $\log(3x)$.

2.12 Exponential and logarithmic equations

9E. $32 = 2^5$, so $x + 1 = 5$ and $x = 4$.

9H. (a) $3^x = 40$, so $x = \dfrac{\ln 40}{\ln 3} \approx 3.36$. (b) $x(x-4) = 5$, so $x^2 - 4x - 5 = 0$ and $x = 5$ or -1. -1 is extraneous (not in the domain), so $x = 5$. (c) $u^2 - 5u + 4 = 0$, so $u = 1$ or 4, giving $x = 0$ or $x = 2$.

2.13–2.14 Log modeling and semi-log plots

10E. $\log(1000I) = \log 1000 + \log I = 3 + \log I$. **3 units** higher.

10H. (a) $\log y \approx 0.699,\ 1.176,\ 1.653,\ 2.130$. (b) Each step adds 0.477 ($= \log 3$): a line with slope 0.477. (c) $\log y = 0.477x + 0.699$. Then $y = 10^{0.699} \cdot (10^{0.477})^x = 5 \cdot 3^x$.